A 46GS/s 7-bit Time-Interleaved Time-Domain ADC with Synthesizable Unit ADCs in 16nm FinFET

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Abstract

A 7-bit 16-way time-interleaved time-domain analog-to-digital converter (TDADC) is presented with unit ADCs synthesized using Verilog code and fully-automated standard digital place-and-route techniques. The proposed design introduces capacitor-based distortion compensation in the voltage-to-time converter and employs a novel 3-input time comparator with implicit phase interpolation in the fine time-to-digital converter to reduce power and area. Fabricated in 16nm FinFET, the proposed ADC operates at 46GS/s, achieves 78fJ/conv.-step at Nyquist, has 20.1GHz 3dB input bandwidth, and occupies 0.085mm² area.

Introduction

Interest in high-speed time-domain ADCs (TDADCs) arises from their superior compatibility with advanced digital CMOS technologies relative to voltage-domain counterparts [1–4]. These ADCs rely on timing signals with digital properties, which enhances integration with standard synthesis flows to enable realization of synthesizable ADCs. However, TDADC performance is often limited by distortion in the voltage-to-time converter (VTC) and time-to-digital converter (TDC) complexity. This work presents a time-interleaved (TI) TDADC with synthesizable unit ADCs that introduce a VTC with MOS capacitor-based third-order nonlinearity compensation and employ a novel 3-input time comparator (I-TCMP) with implicit phase interpolation in the calibration-free fine TDC to reduce the number of phase interpolators (PIs). Automatic layout generation of the unit ADCs is achieved with Verilog code that calls digital standard cells and custom analog unit VTC and I-TCMP cells and employing digital place-and-route (P&R) techniques. These unit ADCs are 16-way interleaved with a high-bandwidth interleaver to achieve 46GS/s operation.

Architecture and Unit TDADC Synthesis Flow

Fig. 1 shows the proposed 46GS/s 16-channel TI TDADC architecture with synthesizable unit ADCs. The input interleaver includes an analog front-end (AFE) with T-coil ESD termination and Rank 1 buffers followed by bootstrapped track-and-holds (T/Hs) and Rank 2 buffers. A clock generation block utilizes an 11.5GHz input clock to provide the required T/H and unit TDADC clocks. While the interleaver and clock generation circuitry are manually designed and routed, automatic layout generation is utilized for the unit TDADCs.

Unit ADC design automation is achieved with the Python wrapper shown in Fig. 2. A fully-digital Verilog code module and custom analog components (VTC and I-TCMP) are combined to create the TDADC netlist, which is then synthesized and routed using standard P&R techniques with timing and area constraints. The generated layout is extracted and Ocean script simulations verify the design against ADC performance metrics, such as the specific stage delays used in the coarse TDC ($T_{R1\text{-}3}$). These delays are iteratively adjusted until all specifications are satisfied by adding or removing standard cell buffers and updating the netlist and layout constraints accordingly.

ADC Circuits

The unit TDADC shown in Fig. 3 is comprised of a VTC, a 4-bit delay-tracking pipelined coarse TDC, and a 3-bit calibration-free flash fine TDC. The differential input is sampled onto two VTC capacitors with a shared common-mode node that is discharged to mitigate current mismatch [1]. Introducing a MOS-capacitor at this discharge node with C-V characteristics that compensate for current source nonlinearity improve the VTC SFDR by 2.7dB in simulation.

Fig. 4 shows the block diagram of the delay tracking pipelined coarse TDC [2]. The resolution of each stage (T_{R1}, T_{R2}, T_{R3}) is determined by the difference between two signal paths, with a time comparator resolving each bit per stage. T_{R1}, T_{R2}, and T_{R3} are sequentially subtracted from the input pulse width to subsequently resolve MSB-1, MSB-2, and MSB-3, respectively. The synthesis flow adjusts the number of buffers to achieve the per-stage target resolution of $T_{FS}/2$, $T_{FS}/4$, and $T_{FS}/8$, where T_{FS} is the nominal 114.28ps full-scale pulse width at the VTC output. Including OR gates in each T_{min} path aligns the output falling edges at each stage to allow for robust quantization with NAND gate-based time comparators. Standard cell-based 6-bit capacitor DACs are used at each stage for fine-tuning and compensating process variations in the delay lines through foreground calibration.

The calibration-free fine flash TDC is shown in Fig. 5. Conventional 3-bit flash structures require at least 19 PI blocks. The required PI blocks in this design are reduced by nearly half with the proposed I-TCMP that evaluates the proximity of the input signal (N) to two references (P1 and P2). A reset mechanism is also employed after generating the four reference phases to mitigate falling edge contention, further reducing power consumption by 35%. Monte Carlo simulations confirm calibration-free fine TDC operation.

Fig. 6 shows the input interleaver and clock generation circuitry. A measured 20.1GHz input bandwidth is achieved by employing high-pass feedback in the Rank 1 buffer and utilizing a speed-enhanced bootstrapped switch T/H [5] that minimizes hold-phase feedthrough using a dummy grounded resistor network. Utilizing a QDLL and an ILO [6], an $f_S/4$ input clock generates 8-phase 25% duty cycle $f_S/8$ $\Phi_{T/H}$ clocks that are distributed to the Rank 1 T/Hs. Skew calibration is performed with 7-bit programmable capacitive loads at the outputs of several buffer stages to ensure consistent edge rates. Finally, 16-phase 25% duty cycle $f_S/16$ Φ_{ADC} clocks, derived from the $f_S/8$ clocks, are distributed to the unit TDADCs.

Experimental Results

Fig. 7 shows the measured ADC SNDR/SFDR versus input frequency, with the SNDR varying from 35.13dB to 33.18dB for low and Nyquist frequencies, respectively. The 32,768-point ADC output spectrums in Fig. 8 show that the implemented calibration circuitry allows the offset and gain/skew spurs to be comparable or below the signal harmonics. Fig. 9 shows the measured DNL ($< \pm 0.8$ LSB) and INL ($+1.02/-0.95$ LSB) and the 16nm die micrograph. The TI TDADC occupies an active area of only 0.085mm² and consumes 133.6mW. Tables I and II provide a detailed comparison of this ADC with other state-of-the-art synthesizable and high sample rate time-interleaved ADCs. Overall, the unit TDADC implementation allows for the fastest channel sample rate amongst all designs and results in a dramatic increase in sample rate amongst the synthesizable designs and the lowest area and competitive FoM amongst the high sample rate designs.

Acknowledgements

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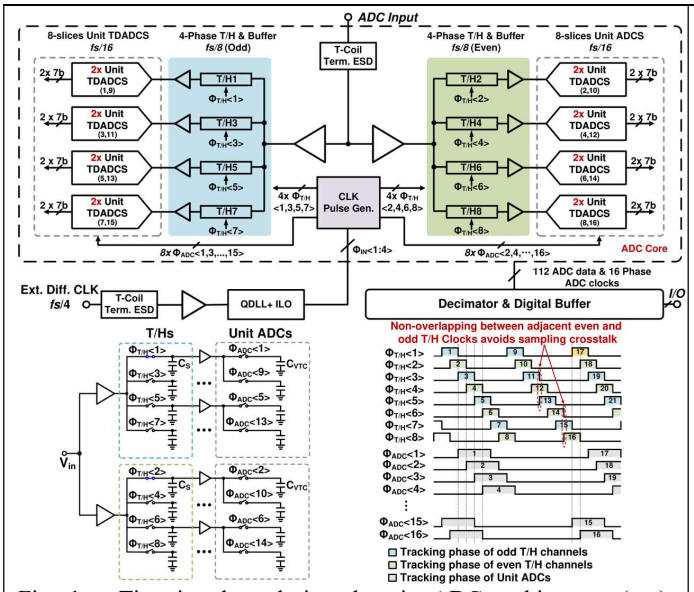


Fig. 1. Time-interleaved time-domain ADC architecture (top), interleaver timing diagram (bottom).

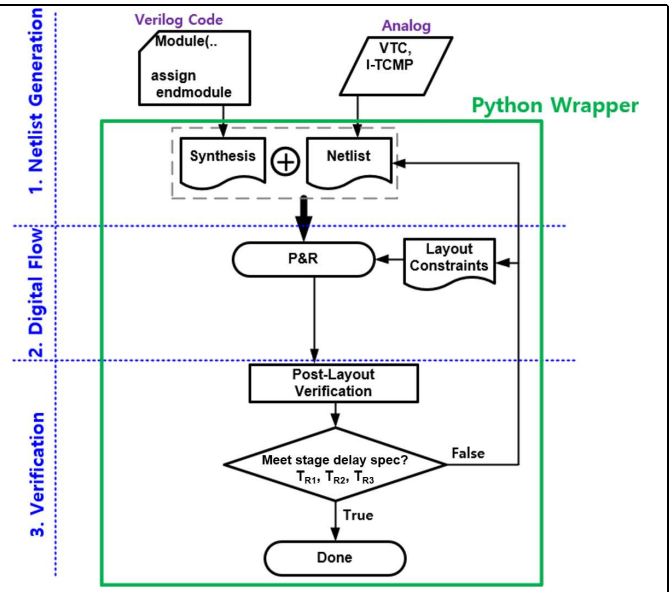


Fig. 2. Unit TDADC design flow that utilizes standard digital synthesis and P&R techniques.

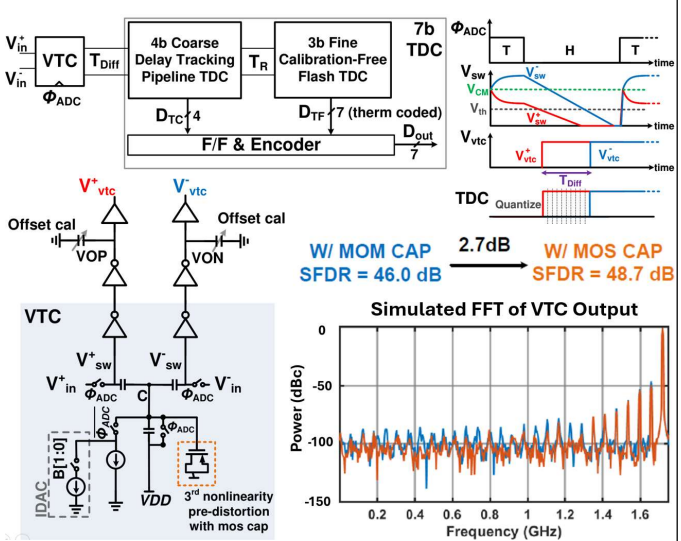


Fig. 3. Automatically-generated unit TDADC block and timing diagrams (top) and VTC with proposed MOS-capacitor-based nonlinearity compensation (bottom).

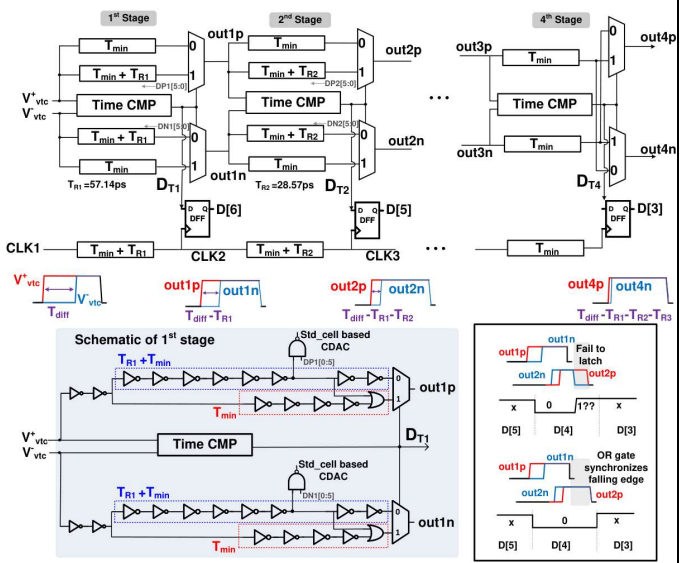


Fig. 4. Coarse TDC block diagram (top), 1st stage schematic (bottom left), and falling edge synchronization (bottom right).

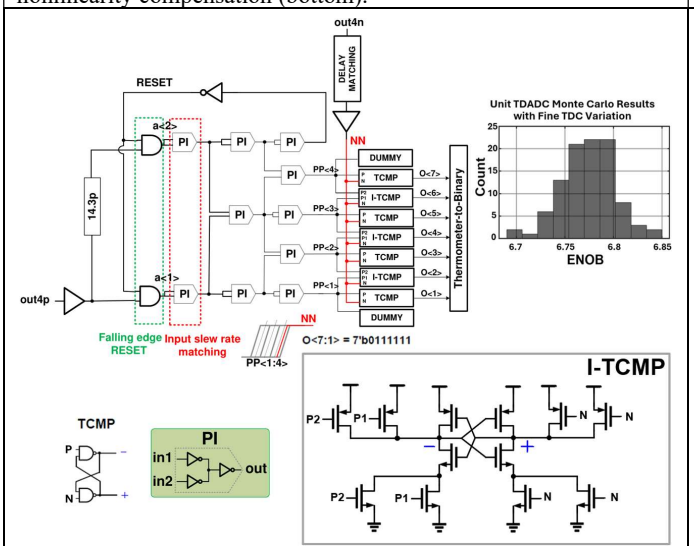


Fig. 5. Calibration-free fine TDC block diagram and Monte Carlo simulations (top). Time Comparator (TCMP), Phase Interpolator (PI), and proposed 3-Input Time Comparator (I-TCMP) with implicit phase interpolation schematics (bottom).

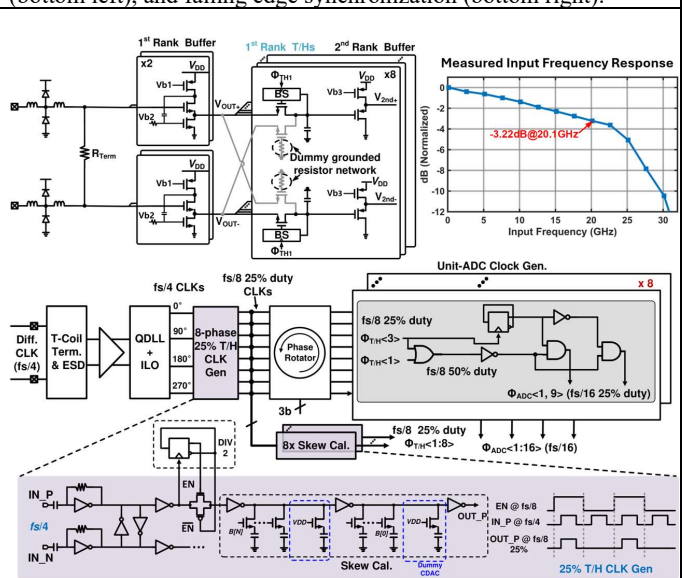


Fig. 6. Input interleaver (top) and clock generation and skew correction circuitry (bottom).

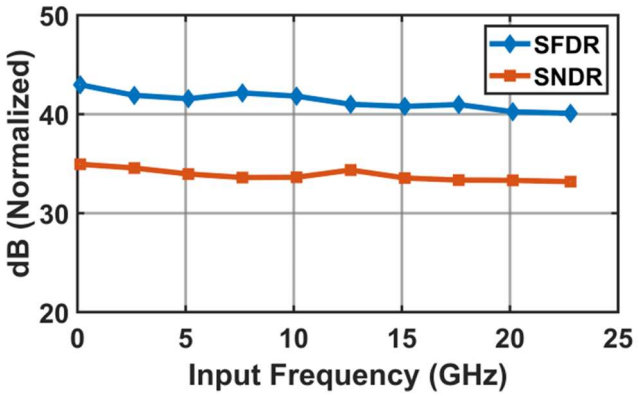


Fig. 7. Measured ADC SNDR and SFDR vs input frequency.

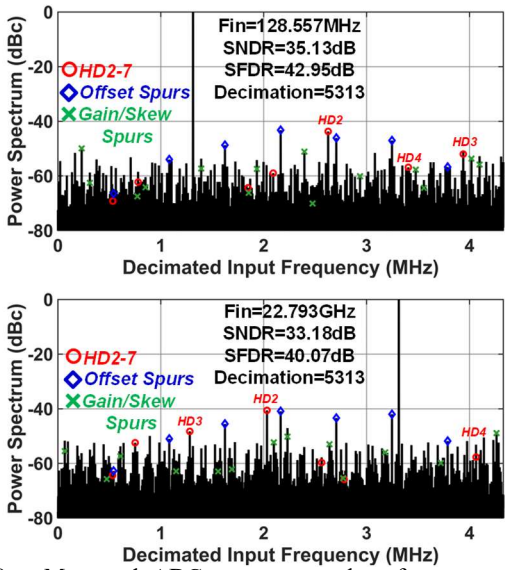


Fig. 8. Measured ADC spectrum at low frequency (top) and Nyquist frequency (bottom).

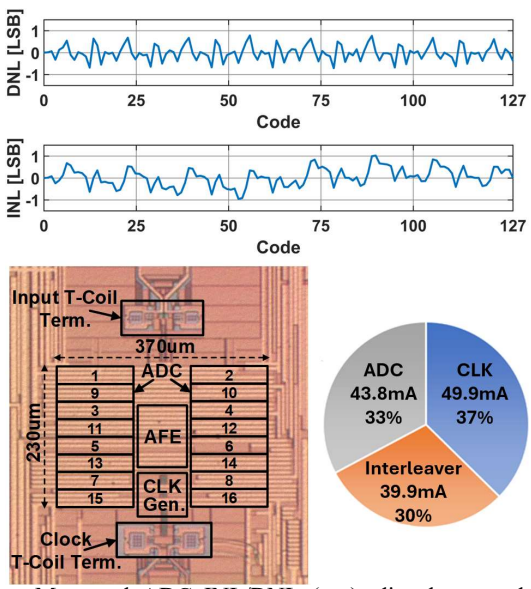


Fig. 9. Measured ADC INL/DNL (top), die photo, and power breakdown (bottom).

TABLE I. Comparison with Synthesizable ADCs

	[7] Kim VLSI '20	[8] Park ACCESS '19	[9] Xu TVLSI '21	[10] Seo TCASII '18	[11] Liu ESSERC '24	This Work
Tech.	16nm	28nm	65nm	28nm	16nm	**16nm**
Resolution (bits)	8	8	6	11	12	**7**
Fs (GS/s)	20	0.0005	0.001	0.05	4	**46**
Power (mW)	175	0.092	0.9	0.399	124.6	**133.6**
Area (mm^2)	0.003	0.002	0.1	0.009	0.36	**0.085**
Channels	16	1	1	1	8	**16**
SNDR @Nyq. (dB)	35.472	34.2	35.7	56.8	56.5	**33.18**
SFDR @Nyq. (dB)	NA	44	45.7	69.2	72.9	**40.07**
FoM (fJ/step)	180	4390	1860	14.1	57.1	**78**

TABLE II. Comparison with Time-Interleaved ADCs

	[12] Kull ISSCC '18	[5] Zhu JSSC '23	[4] A. A. JSSC '23	[13] Nani VLSI '24	[3] A. W. SSCL '24	This Work
Channel arch.	SAR	Pipe. SAR	TDC + SAR	Loop-unrolled SAR	TDC-assisted SAR	**TDC (Pipe. Coarse + Flash Fine)**
Tech.	14nm	22nm	22nm	5nm	22nm	**16nm**
Resolution (bits)	8	7	7	7	7	**7**
Fs (GS/s)	72	38	40	60	50	**46**
Channel Fs (GS/s)	1.125	1.1875	0.83	0.9375	1.042	**2.875**
Power (mW)	235	119.7	104	109.3	116	**133.6**
Area (mm^2)	0.15	0.107	0.138	0.151	1.04	**0.085**
Channels	64	32	48	64	48	**16**
SNDR @Nyq. (dB)	30.4	35.6	30.7	34.3	30.9	**33.18**
SFDR @Nyq. (dB)	N/A	43.8	48.7	43	41	**40.07**
FoM (fJ/step)	120.7	64.05	92.8	42.88	91.2	**78**

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